Integrating Turkish Wordnet KeNet to Princeton WordNet: The Case of One-to-Many Correspondences

Özge Bakay
Department of Foreign Language Education
Boğaziçi University
İstanbul, Turkey
ozge.bakay@boun.edu.tr

Özlem Ergelen
Department of Linguistics
Boğaziçi University
İstanbul, Turkey
ozlem.ergelen@boun.edu.tr

Olcay Taner Yıldız
Department of Computer Engineering
Işık University
İstanbul, Turkey
olcaytaner@isikun.edu.tr

***Abstract*—In this paper, we introduce a novel approach of forming interlingual relations between multilingual wordnets. We have mapped Turkish senses in KeNet with their corresponding senses in Princeton WordNet by drawing one-to-many correspondences. As a result of language-specific properties, one synset in one language is matched with multiple synsets in the other language in some cases. Our method of integrating KeNet into a multilingual network also included mapping the most frequent 5000 senses in English with their equivalent senses in Turkish. What we demonstrate is that one-to-many interlingual correspondances are necessary to include in mappings both from Turkish-to-English and English-to-Turkish. Furthermore, one-to-many mappings give us insights into the semantic relations to be constructed in Turkish, such as hypernymy.**

Keywords*—*Wordnet, KeNet, Interlingual relations

I. Introduction

The idea of a wordnet, which was first suggested by [1], has developed into a comprehensive semantic network between the lexical items of a language. The first wordnet was in English, namely Princeton WordNet (PWN), developed at Princeton University. The main building blocks of a typical wordnet are synsets, which are comprised of lexical items representing a unique sense. Ever since the introduction of PWN, wordnets in various languages have been constructed. Subsequently, multilingual wordnets have been formed in order to incorporate different languages and to utilize them in natural language processing (NLP) and machine translation. Different strategies have been adopted to construct interlingual relations between the wordnets of different languages. In this study, we will present our own strategy to form interlingual relations between KeNet, a comprehensive Turkish wordnet [2], and the English WordNet PWN. We have employed a one-to-many correspondence method to map Turkish synsets with the English ones, in contrast to many other attempts adopting a one-to-one strategy.

The outline of the rest of this paper is as follows: Section II will go over the strategies used in other wordnets to form multilingual networks. Section III will discuss the strategy adopted in this work to form interlingual relations from KeNet to PWN and our solutions to the problems we have encountered. Section III-A will discuss these issues in detail, and Section III-B will go over our method to integrate KeNet into a multilingual network. Section IV will present the results of our study and lastly, Section V will conclude this paper.

II. Literature Review

The first multilingual wordnet we will look at is a bilingual wordnet where the Marathi language is linked to Hindi WordNet in an attempt to construct Indo-WordNet [3]. Using *relation borrowing* as a method, this work has established semantic relations in Marathi WordNet (MWN) through the interlingual mapping of its senses with Hindi WordNet (HWN). Regarding the mapping between the senses, they summarize the cases they have encountered into three main categories. The first case is the existence of a sense in both languages, which is the most common one. In this situation, they basically copy the semantic relations from HWN. The second case is the absence of a Hindi sense in Marathi. For these senses, they do not include the relations. The last case is the one when a sense in Marathi does not exist in Hindi. For this category of senses, they establish the sense relations in MWN manually by using intersection similarity based word sense disambiguation for Hindi senses.

Another example for a multilingual wordnet creation is the free French WordNet (FREWN), which uses the *extend approach* [4]. In this approach, a set of synsets from PWN is translated into the target language, here French, with their semantic database and later, these translated relations are manually checked. As opposed to the *merge approach*, according to which a new wordnet is first constructed based on its monolingual relations and then mapped to the other wordnets, the extend approach is argued to be time saving and less demanding. Another important characteristic of such an interlingual wordnet creation procedure is that it assumes that concepts and semantic relations between the languages are language independent.

A different mapping strategy used in the literature is the manual strategy, which is also adopted to map the Polish WordNet (plWordNet) to PWN [5]. The major difference of this strategy from the previous ones is that plWordNet takes lexical units, rather than synsets, as the basic notion. So, Polish

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Table I. ONE-TO-MANY CORRESPONDENCES FOR ENGLISH

Turkish			English		
Id	Literal(s)	Definition	Id	Literals	Definition
TUR10-0322140	haksız nahak	Hak ve adalete uygun olmayan	ENG31-00961901-a	inequitable unjust	not equitable or fair
			ENG31-00960366-a	unfair unjust	not fair
			ENG31-02044679-a	unrighteous	not righteous
			ENG31-01951967-s	undue unjustified unwarranted	lacking justification or authorization
			ENG31-01411393-s	unlawful wrongful	having no legally established claim

Table II. LANGUAGE-SPECIFIC CLASSIFICATION

Turkish			English		
Id	Literal(s)	Definition	Id	Literals	Definition
TUR10-0396530	kadın bayan	Kadınların ad veya soyadlarının önüne getirilen saygı sözü	ENG31-06352619-n	Miss	a form of address for an unmarried woman
			ENG31-06352801-n	Mrs Mrs.	a form of address for a married woman
			ENG31-06352895-n	Ms Ms.	a form of address for a woman

lexical units are matched to PWN synsets by means of their semantic relations. During the mapping process, the editor's first job is to become familiar with the sense of a source language, here Polish. After this, the editor looks for candidate PWN synset(s) to link the Polish synset with, and lastly, selects the PWN synset and the appropriate interlingual relation. The order in which relations are applied is synonymy, hyponymy, hypernymy, meronymy, holonymy, near-synonymy and inter-register synonymy. Regarding the semantic relations, after the selection of the top-most relations, the rest at the bottom is not looked for any more.

As for Turkish, the first comprehensive Turkish wordnet is BalkaNet, where the semantic relations are taken from a monolingual Turkish dictionary [6]. In order to integrate Turkish wordnet of BalkaNet (henceforth, TR-wordnet of BalkaNet) into other multilingual wordnets, the semantic senses are taken from PWN and matched one-to-one from Turkish to English. In BalkaNet and some other multilingual WordNets such as EuroWordNet (EWN) [7], an interlingual index (ILI) is used for one-to-one correspondences to PWN in order to represent concepts that are language-specific [8]. ILI is a principal tool to mapp interlingual relations and to build semantic networks in these wordnets. One drawback of such a matching pattern is that the sense disambiguation made in English is directly translated into Turkish, without considering whether such a sense disambiguation is observable in Turkish or not. In order to eliminate such adverse effects of one-to-one matching, in our work, we allow one-to-many correspondences when it is required.

III. INTERLINGUAL RELATIONS

In this study, we use a strategy similar to *the extend approach* adopted by [4] and display the candidates of interlingual relations between English and Turkish, which have been extracted based on the translations of the synsets, on Google Sheets. These interlingual matching candidates are then checked manually to decide whether they form interlingual relations. Our primary goal in this work is to map the Turkish synsets with their English equivalents. Additionally, an extra advantage of such a mapping strategy is that it has enabled us to better analyze and group the current Turkish synsets and improve KeNet. In the following sections, we will look at the cases we have encountered in the multilingual mapping process in further detail.

A. One-to-many correspondences

When we attempt to match Turkish items with those of English, we have observed that in some cases, it is not possible to select a single synset in English that perfectly matches the semantic sense of the Turkish synset. Our principal attempt was to minimize the matchings and leave only one-to-one matchings. However, such one-to-one correspondence is not usually possible as it leaves one side of the mapping incomplete or even false. To exemplify the case of one-to-many matchings from English to Turkish, kinship terms are more varied in Turkish than in English. This leads to multiple matchings of Turkish kinship terms with those of English. The terms teyze "mother's sister" and hala "father's sister" can be translated into one single term, "aunt" in English, which does not carry the distinction that Turkish has. It is also often the case that multiple English words correspond to a single word in Turkish. For example, the terms "information" and "knowledge" in English can both be translated into Turkish as "bilgi", often without reflecting the distinction made in English. As a result, the inability to have exact interlingual matchings for particular items has given rise to one-to-many correspondences in our interlingual data. Section III-A1 will discuss data from English and Section III-A2 from Turkish.

1) English: We have realized that some distinctions that are made in the English WordNet do not have any correspondence in KeNet. This has given rise to one-to many correspondences from Turkish to English. That is, one synset in Turkish is matched with more than one synset in English as Turkish does not have the distinction in English. One example is given in Table I. Note that the part of speech (POS) of the group of synsets in the table is adjective and the synsets in English do not have a hypernymy relation to one another. Given that we were unable to find a single synset in English which corresponds to the Turkish synset "haksız, nahak", we mapped this synset with five different synsets in English. Hence, all English synsets in the table were matched with a single synset in Turkish.

Table III. MULTIPLE MATCHING FROM ENGLISH TO TURKISH

Turkish			English		
Id	Literal(s)	Definition	Id	Literals	Definition
TUR10-0320470	hafiflemek	Herhangi bir sebeple eski ağırlığı azalmak	ENG31-00245945-v	abate let up slack off slack die away	become less in amount or intensity
TUR10-0320480	hafiflemek azalmak	Etkisi, gücü azalmak			
TUR10-0451010	kırılmak yatışmak	Soğuk, rüzgâr vb. eski gücü kalmamak, azalmak			

Table IV. CAUSATIVITY IN TURKISH

Turkish			English		
Id	Literal(s)	Definition	Id	Literals	Definition
TUR10-0434330	kaynamak haşlanmak	Yiyecek, içecek pişmek	ENG31-00328938-v	boil	immerse or be immersed in a boiling liquid, often for cooking purposes
TUR10-0434850	kaynatmak	Kaynamasını sağlamak			

One reason for the occurence of one-to-many correspondences is that the categorizations and classifications that are made in English are not found in Turkish such as gender marking, the formal-informal speech differentiation etc. One such example is the distinction made in addressing a woman in English based on her marital status, which is shown in Table II. In Table II, the items “Miss”, “Mrs” and “Ms”, the forms of address for women, is represented by a single sense in Turkish, which is “kadın, bayan”. Since Turkish does not make a distinction in addressing a married and an unmarried woman, we matched a single Turkish synset, “kadın, bayan”, with three different synsets in English.

2) Turkish: One-to-many matchings have also occurred when items are mapped from English to Turkish. These matchings can be grouped into three: (i) one English synset matching with multiple Turkish synsets, (ii) the expression of a concept through Turkish overt morphological markings and (iii) problems in the organization of Turkish synsets. Regarding the first case, we have observed that one English synset might be matched with multiple different synsets in Turkish. The reason behind that is the lack of the sense distinction provided by Turkish synsets in English. One example of such a sense distinction is given in Table III, where three different synset in Turkish correspond to only one synset in English. The synsets "hafiflemek" (to lose strength), "hafiflemek, azalmak" (to lose strength or effect) and "kırılmak, yatışmak" (to lose strength, to become weaker (for cold, wind etc.)) correspond to the same English synset, "abate, let up, slack off, slack, die away", as shown in Table III.

a) Morphological correspondence: causative: Mapping from English to Turkish has also caused one-to-many correspondences due to the morphological marking of certain syntactic functions in Turkish. The syntactic differentiation provided through morphological markings in Turkish cannot always be reflected in English. Such differentiations which are usually made through suffixes in Turkish are often expressed by a separate word in English. In interlingual mapping, this has led to one-to-many correspondences from English to Turkish with respect to causativity (This would also be the case for passivized verbs in Turkish as they denote different morphological marking; however, passivized forms of verbs are not included as separate synsets in KeNet). In the example provided in Table IV, the English sense “boil” is matched with two different synsets in Turkish, "kaynamak" (to boil food or liquid) and "kaynatmak" (to cause to boil). The second lexical item possesses a causative ending -t attached to the verb "kayna-" (boil). This causativity differentiation made in Turkish lexical items through a bound morphological marking is not made in English since English does not possess such a morphological marker. The description of the item “boil”, which includes “immerse or be immersed”, clearly demonstrates that it covers both senses which are disambiguated in Turkish.

b) Merging of Turkish synsets: English-to-Turkish multiple correspondences have displayed certain instances where the corresponding Turkish synsets need to be merged. As mentioned earlier, such a display of interlingual mapping has enabled us to realize the cases where there are multiple synsets in Turkish referring to the same synset in English. Whereas this might reflect a one-to-many correspondance from English to Turkish, as discussed in III-A2, it has also showed cases where the observable one-to-many correspondence is only due to the lack of the necessary synset merging in Turkish. One example is given in Table V, in which two different Turkish synsets are matched with a single synset in English. In this example, the definition of the first synset in Turkish given in Table V is marked as “null” and only the second synset has a dictionary definition. When these two synsets are searched in a Turkish dictionary, it is found out that they are actually synonyms and hence, should be in the same synset. Therefore, in such cases, we have concluded that the two synsets in Turkish should be merged into one single synset. This kind of mergings has been useful for a better organization of the synonymy relation in KeNet synsets (see [9] for a detailed discussion of synset construction in KeNet).

As a further note on this merging case, the data in Table VI presents a merging case. In this example, the definition of the second synset, "uyandırmak", exists as a separate Turkish synset, as shown in the upper line in VI. So, such cases where the definition of a given synset is comprised of a single word have been checked for possible merging, which has again contributed to the synset organization of KeNet.

3) Hypernymy relation: We must also note that the interlingual matchings we have made in this study feed into the hypernymy relations in KeNet. To exemplify, a Turkish synset might be matched with two English synsets that have a hypernymy-hyponymy relation in English. This means that the hypernymy relation that holds for English is lost in Turkish. In the example in Table VII, the Turkish synset "giriş, methal" corresponds to four different English synsets. Regarding hypernymy, on the English side, English Synset numbered 1 is a

Table V. MERGING OF TURKISH SYNSETS

Turkish			English		
Id	Literal(s)	Definition	Id	Literals	Definition
TUR10-0701410	solumak nefes alıp vermek		ENG31-00001740-v	breathe take a breath respire suspire	draw air into, and expel out of, the lungs
TUR10-0902510	içine çekmek teneffüs etmek	soluk almak			

Table VI. MERGING OF TURKISH SYNSETS BASED ON CAUSATIVE MARKING

Turkish			English		
Id	Literal(s)	Definition	Id	Literals	Definition
TUR10-0801290	uyandırmak	Uyanmasına yol açmak	ENG31-00018806-v	awaken wake waken rouse wake up arouse	cause to become awake or conscious
TUR10-0801840	uyarmak kaldırmak	Uyandırmak			

Table VII. HYPERNYMY RELATION

Turkish			English		
Id	Literal(s)	Definition	Id	Literals	Definition
TUR10-0298370	giriş methal	Bir yapıda içeri geçilen yer, methal, antre	ENG31-03295682-n	entrance entranceway entryway entry entree	something that provides access (to get in or get out)
			ENG31-02718424-n	anteroom antechamber entrance hall hall foyer lobby vestibule	a large entrance or reception room or area
			ENG31-03433387-n	gateway	an entrance that can be closed by a gate
			ENG31-03228735-n	doorway door room access threshold	the entrance (the space in a wall) through which you enter or leave a room or building

hypernym of English Synsets numbered 3 and 4. In Turkish, all these three synsets are matched with a single synset, which has caused the hypernymy relation between the English synsets to remain uncovered in Turkish. Such cases demonstrate that one-by-one interlingual mapping of synsets is unable to cover the discrepancies between the languages with respect to their semantic relations, as well.

B. KeNet and Multilingual WordNet: Top 5000 words

As a complementary part of our aim to integrate KeNet into a multilingual wordnet and to cover the most-commonly-used overlapping senses in languages, we have extracted 5000 most frequent English senses from PWN and matched them with their Turkish counterparts. First, we have left out the synsets that have no correspondence in Turkish, such as "able" in English, as they do not have a lexical counterpart in Turkish. Then, we have matched the remaining English synsets with the ones in Turkish. We have also excluded the Turkish synsets that do not translate into English. As a result of this study, we have matched 4417 of these 5000 common lexical items in English with their Turkish correspondents (88,34%).

IV. RESULTS

In this work, we started with 15333 lines of English-Turkish synset matchings. We deleted those that were not found to have a match. At the end of the study, we are left with 4765 lines of English-Turkish correspondences. 846 of these are one-to-many correspondences from Turkish to English and 411 from English-to-Turkish. Furthermore, we put aside 152 Turkish synsets for further splitting process since these synsets contained items that are not in a true synonymy relation. After the splitting process for those items, we have completed the interlingual mapping of the newly-created synsets, as well. Additionally, 30 synsets in Turkish that were either marked as null, meaning that there is no definition, or included a code instead of a real definition in the definition section have been revealed. Correction of these synsets have contributed to the improvement of KeNet synset organization.

V. CONCLUSION

In this paper, we have discussed the methods we have adopted in order to map the Turkish wordnet KeNet to the English WordNet PWN. We have mentioned some previous approaches to develop a multilingual semantic network and presented our own strategy of Turkish-to-English mapping and integrating KeNet into a multilingual wordnet. We have adopted a one-to-many matching method since in some cases, the sense disambiguation made in English is not relevant for Turkish, and vice versa. In other words, one synset in one language has been matched with more than one synset in the other language in our work to better capture the interlingual relations between Turkish and English. Moreover, for a better representation of Turkish with more overlapping senses among languages, we have extracted the most frequent 5000 English synsets and matched them with compatible Turkish synsets. Two further advantages of our approach are that we were able to (i) improve the organization of the KeNet synsets and (ii) reveal the discrepancies between English and Turkish in terms of the hypernymy relation, which would otherwise go unnoticed in a one-to-one matching strategy.

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